

Customer Behavior Analysis

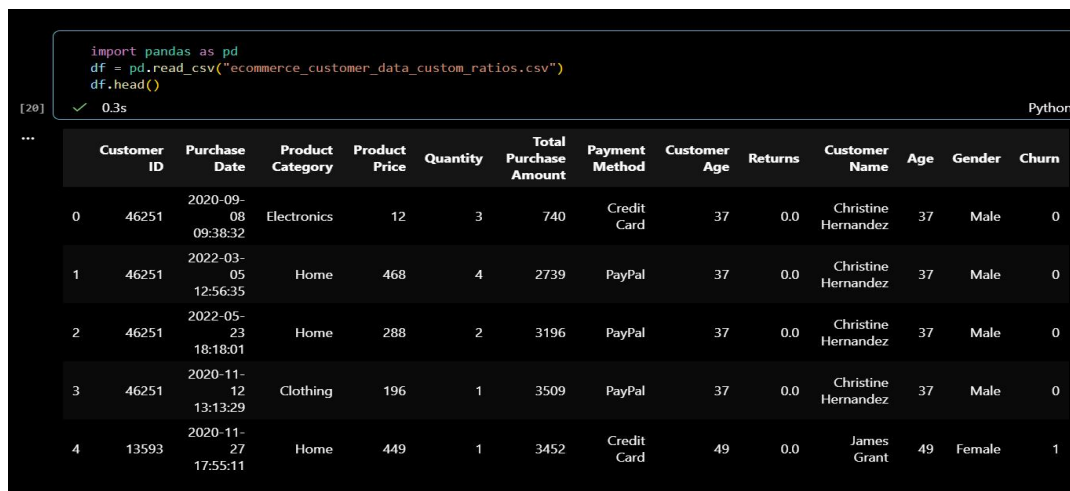
1. Objective

The objective of this project is to analyze customer purchasing behavior, identify trends, and generate business insights using Python and Tableau.

2. Dataset Information

Attribute	Details
Dataset Name	Ecommerce Customer Data
Table Records	250,000
Table Columns	13
Tools Used	Python(Pandas),Tableau
File Format	CSV

3. Data Cleaning



The screenshot shows a Jupyter Notebook interface. At the top, a code cell contains the following Python code:

```
import pandas as pd
df = pd.read_csv("ecommerce_customer_data_custom_ratios.csv")
df.head()
```

 Below the code cell, the output is displayed as a table preview. The table has 13 columns: Customer ID, Purchase Date, Product Category, Product Price, Quantity, Total Purchase Amount, Payment Method, Customer Age, Returns, Customer Name, Age, Gender, and Churn. The first five rows of data are shown, with the first four rows belonging to Customer ID 46251 and the fifth row belonging to Customer ID 13593.

	Customer ID	Purchase Date	Product Category	Product Price	Quantity	Total Purchase Amount	Payment Method	Customer Age	Returns	Customer Name	Age	Gender	Churn
0	46251	2020-09-08 09:38:32	Electronics	12	3	740	Credit Card	37	0.0	Christine Hernandez	37	Male	0
1	46251	2022-03-05 12:56:35	Home	468	4	2739	PayPal	37	0.0	Christine Hernandez	37	Male	0
2	46251	2022-05-23 18:18:01	Home	288	2	3196	PayPal	37	0.0	Christine Hernandez	37	Male	0
3	46251	2020-11-12 13:13:29	Clothing	196	1	3509	PayPal	37	0.0	Christine Hernandez	37	Male	0
4	13593	2020-11-27 17:55:11	Home	449	1	3452	Credit Card	49	0.0	James Grant	49	Female	1

Figure 1:Dataset Preview

Observation: The dataset contains customer information such as purchase date, product category, quantity, payment method, customer age, gender, and churn status.

```

df.info()
[21] ✓ 0.0s Python

...
<class 'pandas.DataFrame'>
RangeIndex: 250000 entries, 0 to 249999
Data columns (total 13 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Customer ID            250000 non-null  int64
1   Purchase Date          250000 non-null  str
2   Product Category       250000 non-null  str
3   Product Price          250000 non-null  int64
4   Quantity               250000 non-null  int64
5   Total Purchase Amount  250000 non-null  int64
6   Payment Method         250000 non-null  str
7   Customer Age           250000 non-null  int64
8   Returns                202404 non-null  float64
9   Customer Name          250000 non-null  str
10  Age                    250000 non-null  int64
11  Gender                 250000 non-null  str
12  Churn                  250000 non-null  int64
dtypes: float64(1), int64(7), str(5)
memory usage: 24.8 MB

```

Figure 2: Dataset Information

Observation: The dataset contains 250,000 records and 13 columns. Most columns have complete data, while the **Returns** column contains missing values.

```

df.isnull().sum()
[22] ✓ 0.0s Python

...
Customer ID            0
Purchase Date          0
Product Category       0
Product Price          0
Quantity               0
Total Purchase Amount  0
Payment Method         0
Customer Age           0
Returns                47596
Customer Name          0
Age                    0
Gender                 0
Churn                  0
dtype: int64

```

Figure 3: Missing Values

Observation: Only the **Returns** column has missing values (47,596). All other columns are complete.

4. Summary Statistics

```

df.mean(numeric_only=True)
[23] ✓ 0.0s

...
Customer ID            25004.036240
Product Price          254.659512
Quantity               2.998896
Total Purchase Amount  2725.370732
Customer Age           43.940528
Returns                0.497861
Age                    43.940528
Churn                  0.199496
dtype: float64

```

Figure 4: Mean Output

```

df.median(numeric_only=True)
[24] ✓ 0.0s

...
Customer ID            25018.0
Product Price          255.0
Quantity               3.0
Total Purchase Amount  2724.0
Customer Age           44.0
Returns                0.0
Age                    44.0
Churn                  0.0
dtype: float64

```

Figure 5: Median Output

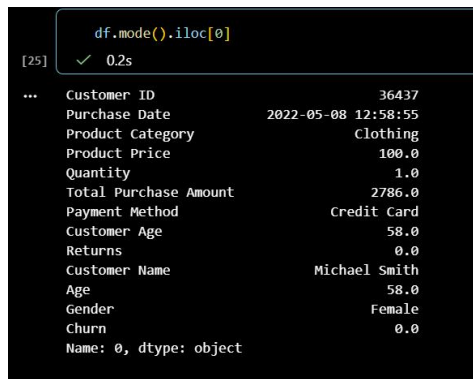


Figure 6: Mode Output

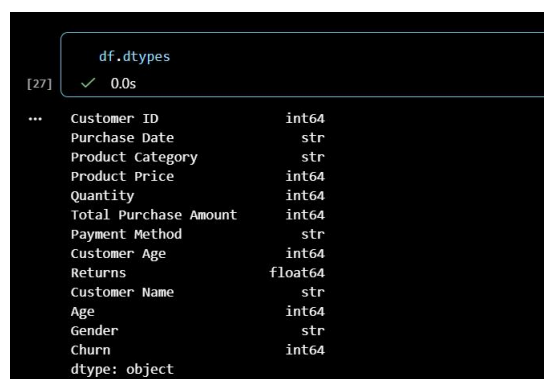


Figure 7: Data Types

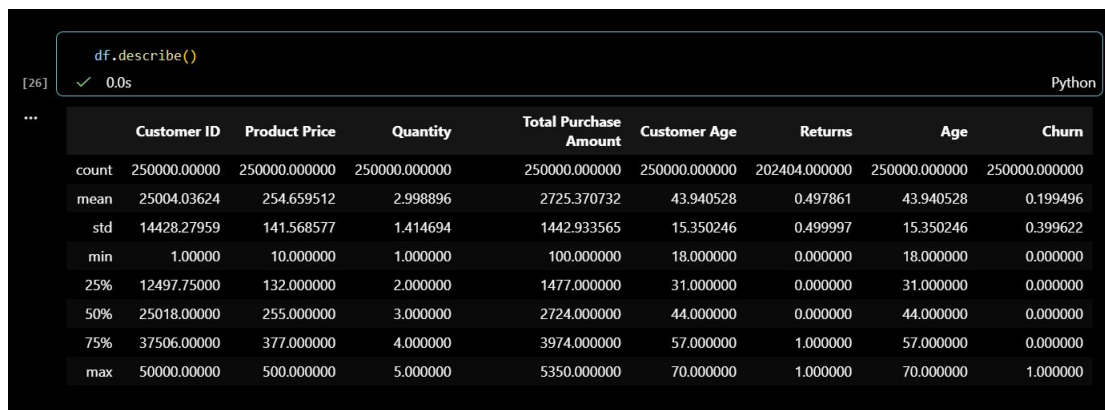


Figure 8: Describe Output

Observation: The average purchase amount is approximately **2725**, the average customer age is **44 years**, and customers purchase around **3 products** per transaction on average.

5. Visualizations

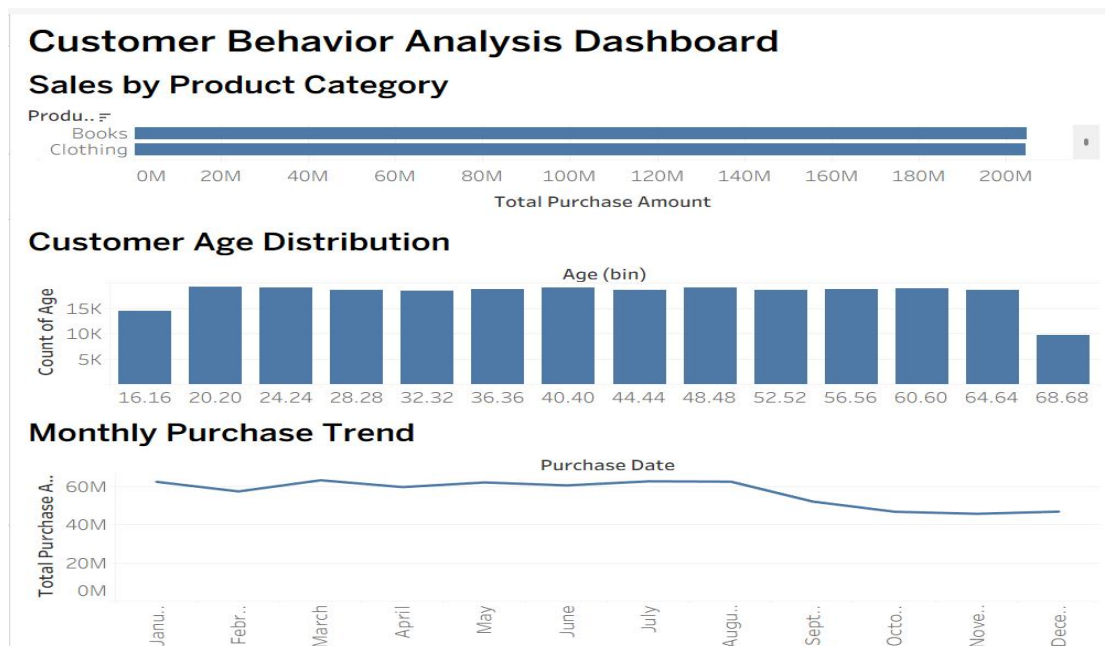


Figure 9: Dashboard

Sales by Product Category:

Books and Clothing generated the highest sales, while Electronics generated comparatively lower revenue.

Customer Age Distribution:

Customers are distributed across a wide age range, with the highest concentration in the adult age groups.

Monthly Purchase Trend:

Sales remain relatively stable throughout the year, with slight fluctuations between months.

6. Executive Summary

This project analyzed customer purchasing behavior using Python and Tableau. Exploratory Data Analysis (EDA) was performed to understand customer demographics, purchasing trends, and sales performance. The results indicate stable purchasing behavior throughout the year, with Books and Clothing contributing the highest revenue. The dashboard provides an easy-to-understand visualization of key business metrics that can support data-driven decision-making.

7. Business Insights

1. Books and Clothing are the top-performing product categories and should receive more promotional focus.
2. Most customers belong to the adult age group, making them an important target audience for marketing campaigns.
3. Sales remain consistent across months, enabling effective inventory and resource planning.